

Module 2: Foundations in chemistry

This module acts as an important bridge into AS and A Level Chemistry from the study of chemistry within science courses at GCSE level.

This module provides learners with a knowledge and understanding of the important chemical ideas that underpin the study of A Level Chemistry:

- atomic structure
- quantitative chemistry: formulae, equations, amount of substance and the mole
- reactions of acids
- oxidation number and redox reactions
- bonding and structure.

The importance of these basic chemical concepts is seen as a prerequisite for all further chemistry modules, and it is recommended that this module should be studied first during this course.

This module allows learners to develop important quantitative techniques involved in measuring masses, gas and solution volumes, including use of volumetric apparatus.

Learners are also able to develop their mathematical skills during their study of amount of substance and when carrying out quantitative practical work.

2.1 Atoms and reactions

This section builds directly from GCSE Science, starting with basic atomic structure and isotopes.

Important basic chemical skills are developed: writing chemical formulae, constructing equations and calculating chemical quantities using the concept of amount of substance.

The role of acids, bases and salts in chemistry is developed in the context of neutralisation reactions.

Finally, redox reactions are studied within the context of oxidation number and electron transfer.

2.1.1 Atomic structure and isotopes

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
Atomic structure and isotopes	
(a) isotopes as atoms of the same element with different numbers of neutrons and different masses	
(b) atomic structure in terms of the numbers of protons, neutrons and electrons for atoms and ions, given the atomic number, mass number and any ionic charge	HSW1 Different models for atomic structure can be used to explain different phenomena, e.g. the Bohr model explains periodic properties. HSW7 The changing accepted models of atomic structure over time. The use of evidence to accept or reject particular models.

Relative mass

- (c) explanation of the terms *relative isotopic mass* (mass compared with 1/12th mass of carbon-12) and *relative atomic mass* (weighted mean mass compared with 1/12th mass of carbon-12), based on the mass of a ^{12}C atom, the standard for atomic masses
- Definitions required.
- (d) use of mass spectrometry in:
- (i) the determination of relative isotopic masses and relative abundances of the isotope,
- (ii) calculation of the relative atomic mass of an element from the relative abundances of its isotopes
- M0.2, M1.2, M3.1*
- Knowledge of the mass spectrometer **not** required.
- Limited to ions with single charges.
- (e) use of the terms *relative molecular mass*, M_r , and *relative formula mass* and their calculation from relative atomic masses.
- For simple molecules, the term *relative molecular mass* will be used.
- For compounds with giant structures, the term *relative formula mass* will be used.
- Definitions of relative molecular mass and relative formula mass will **not** be required.